

Power Supply Design

Linear vs Switching • LDO • Buck • Boost • Layout

The single most-failed-on subsystem of every PCB

Audience: Beginner → Intermediate Engineering Students

Topologies • Component sizing • Layout rules • Protection • Selection

Why Power Supply Matters

Every system needs clean, reliable, efficient power

Foundation

Every digital, analog, RF subsystem starts here.

Noise source

Bad PSU → noisy clocks, ADC jitter, RF spurs.

Efficiency

Wasted W = heat.
Battery life depends on this.

Cost

30-40% of BOM in many products is power-related.

Failure mode

Most field failures trace to power supply faults.

Compliance

EMC, safety, efficiency rating all governed here.

Linear vs Switching Regulators

Pick the right one for the job

Linear (LDO)

- ▶ Drops voltage by dissipating heat
- ▶ Quiet — low noise & ripple
- ▶ Simple: IC + 2 caps
- ▶ Inefficient: $\eta = V_{out}/V_{in}$
- ▶ Best for low ΔV , low current
- ▶ Examples: AMS1117, LP5907, LT3045

Switching (SMPS)

- ▶ Stores energy in inductor/cap
- ▶ Switches at 100 kHz – 5 MHz
- ▶ Efficient ($\eta = 85\text{--}95\%$)
- ▶ Noisy — needs careful layout
- ▶ Complex: inductor + cap + diode + IC
- ▶ Examples: TPS5430, LMR16006, MP2307

Switching Topologies

Buck, boost, buck-boost, flyback, forward

Buck (step-down)

$V_{in} > V_{out}$
Most common in MCUs.

Boost (step-up)

$V_{in} < V_{out}$
Li-ion $\rightarrow$ 5 V USB.

Buck-Boost

V_{in} can be either side
of V_{out} . Used in battery.

SEPIC / Ćuk

Buck-boost with isolation
and low ripple.

Flyback

Isolated, transformer
 $\rightarrow$ off-line PSUs.

Forward / Push-Pull

High-power isolated.
Server PSUs, datacentre.

Buck Converter — Anatomy

The most common switching topology

- ▶ $V_{in} \rightarrow$ high-side MOSFET $\rightarrow$ switching node $\rightarrow$ inductor $\rightarrow V_{out}$
- ▶ Low-side MOSFET (or Schottky) provides current path when high-side is OFF
- ▶ Inductor stores energy; output cap smooths ripple
- ▶ Feedback resistor divider sets V_{out}
- ▶ Control loop adjusts duty cycle = V_{out}/V_{in}
- ▶ Switching frequency: 500 kHz – 2 MHz typical
- ▶ Modern ICs integrate FETs and compensation — "all-in-one"

Key components

Inductor: 4.7–22 μH typical for 1 A class
Input cap: low-ESR 10 μF MLCC
Output cap: 22–47 μF MLCC + bulk
Feedback divider: typ 10 k Ω + 3.3 k Ω

Selection criteria

1. Max output current
2. V_{in} range
3. Switching frequency
4. Efficiency target
5. Quiescent current (battery!)
6. Package & thermal

Component Selection

Inductor • Capacitors • Schottky • Feedback resistors

Component	Selection rule	Common values
Inductor	$L = (V_{in} - V_{out}) \cdot V_{out} / (V_{in} \cdot f_{sw} \cdot \Delta I_L)$; $\Delta I_L \approx 30\%$ of I_{out}	2.2–22 μ H, $I_{sat} > 1.3 \times I_{out}$
Output cap	$C \geq \Delta I_L / (8 \cdot f_{sw} \cdot \Delta V_{out})$; low-ESR MLCC X7R	22–100 μ F
Input cap	Handle pulsed current $\geq I_{out} / 2$	10–22 μ F MLCC + bulk
Schottky	V_f low; $V_{rev} > V_{in} + 30\%$; $I_f > I_{out}$	SS14, SS34 for 1–3 A
Feedback R	Set divider for V_{out} ; balance noise & efficiency	10k top, calc bottom
Compensation	Usually integrated; otherwise per datasheet	Cap + R network

Tip

Use the IC vendor's WEBENCH / LTpowerCAD / PowerNESS tool — it sizes every component automatically.

Switching Regulator Layout

Where most beginner SMPS designs fail

- ▶ Keep switching loop (Vin cap → high-FET → low-FET → Vin cap GND) SMALL
- ▶ Place input cap closest to the IC's VIN pin — minimise loop area
- ▶ Inductor next to switch node — keep switch node trace SHORT
- ▶ Output cap directly after the inductor
- ▶ Solid ground plane underneath the regulator — no splits
- ▶ Sense feedback trace separately, away from switch node
- ▶ Thermal pad of IC has multiple vias to ground plane
- ▶ Snubber (R+C across SW node to GND) damps ringing if needed

LDO Layout & Application

Simpler than SMPS — but stability is non-obvious

- ▶ Input cap close to VIN pin (10 μ F MLCC + bulk)
- ▶ Output cap close to VOUT pin (datasheet-specific)
- ▶ Some LDOs are unstable with ceramic caps — check datasheet ESR window
- ▶ Bypass cap on adjustment pin for low-noise designs
- ▶ Thermal pad on the IC needs copper area = ($P_{diss} \times \theta_{JA}$)
- ▶ Watch input $\rightarrow$ output ratio (large drop = lots of heat)

Stability gotcha

Newer ceramic caps have very low ESR (~ 5 m Ω). Older LDOs were stable only with $ESR \geq 100$ m Ω . Check datasheet stability curve.

PSRR matters

For analog/RF: pick LDO with PSRR > 60 dB at your noise frequency. LP5907 / LT3042 are gold standards.

Boost Converter

Step-up converter for battery → 5 V or higher rails

- ▶ Topology: inductor charges from V_{in} ; switch chops; energy dumps to C_{out} via diode
- ▶ Duty cycle $D = 1 - V_{in}/V_{out}$
- ▶ Inductor current = $I_{out}/(1-D)$ — much higher than buck for same output
- ▶ Output cap sees the SWITCH RIPPLE current — needs low ESR, high ripple rating
- ▶ Common ICs: TPS61023, LMR62421, MP3414, MAX17225 (sync), MT3608 (boost-only)
- ▶ Use cases: Li-ion 3.7 V → 5 V USB; alkaline 1.5 V → 3.3 V; flash for cameras
- ▶ Layout: high-current loop is INPUT → inductor → switch → ground

Power Supply Protection

Layers of defence between mains and your IC

Fuse

PTC (resettable) or fast-blow for over-current.

TVS / MOV

Clamps ESD & lightning surges.

Polarity prot

P-MOSFET or diode against reversed input.

UVLO

Under-voltage lockout — turn off before brownout.

OVLO

Over-voltage lockout — disconnect before damage.

Thermal

Thermal shutdown built into most modern ICs.

Efficiency, Heat & Sizing

Where the lost watts go

Item	Loss mechanism	How to reduce
FET conduction	$I^2 \cdot R_{DS(on)} \cdot \text{duty}$	Pick low- $R_{DS(on)}$ FET / package
FET switching	$\frac{1}{2} \cdot V \cdot I \cdot t \cdot f_{sw}$	Lower f_{sw} , hard-switch fast
Inductor	DCR + core loss	Bigger inductor, better core material
Diode (async)	$V_f \cdot I \cdot (1-D)$	Schottky, low- V_f rectifier
Output cap ESR	$I_{ripple}^2 \cdot ESR$	Multiple parallel MLCC / polymer caps
Quiescent	$I_Q \cdot V_{in}$ always on	Low- I_Q chip for battery products

Sizing rule of thumb

If efficiency at full load is 90% and you draw 5 W, you dissipate 0.55 W — needs ~5 cm² copper pour around the IC for reasonable θ_{JA} .

Common Power Supply Mistakes

What design reviews catch first

Big switching loop

SW node traces long → radiates EMI + voltage spikes.

Split ground plane

Return path forced to detour → noise, EMI.

Wrong inductor

Saturating below I_{out} — converter collapses.

LDO unstable

Wrong cap ESR — oscillates.

Underrated cap

Voltage de-rating ignored — caps degrade fast.

No bulk cap

Just MLCC → input voltage sags under load.

Selection Cheatsheet

Pick the right regulator class

Use case	Topology	Example IC	Notes
Always-on MCU (battery)	Buck, low-IQ	TPS62840, MP2155	μA quiescent current
Sensor analog supply	LDO	LP5907, LT3045	Low-noise, low PSRR
USB → 3.3 V	Buck	TPS62130, MP2307	1–3 A typical
12 V → 5 V (industrial)	Buck	LM2596, LMR16006	Robust, wide Vin
Li-ion → 5 V boost	Boost / SEPIC	TPS61023, MAX17225	Watch peak inductor I
AC-DC isolated	Flyback	TNY278PN, LNK304	Off-the-shelf is safer
High-power DC-DC	Buck or fullbridge	LM5145, LM5160	Watch thermal

Vendor Design Tools

Don't design SMPS by hand — use the tools

TI WEBENCH

Online — pick IC, enter specs, get complete BOM + sim.

LTpowerCAD

Analog Devices / LT — desktop tool, best for performance tuning.

ST eDesignSuite

ST — interactive selection & thermal calculation.

Monolithic PowerNESS

MPS — fast quotation, good for mid-power designs.

Infineon DC-DC

Infineon — large parts library, good for industrial.

Vendor reference designs

Eval boards on every IC page.
Buy one for ₹2-5k, prove the design.

Best Practices Recap

12 habits for power supply design

- ▶ Define ALL rails + currents before picking parts
- ▶ Use vendor tool to size components
- ▶ Buy the eval board first — prove the design
- ▶ Layout: input cap closest to IC, smallest SW loop
- ▶ Continuous ground plane underneath converter
- ▶ Thermal pad with multiple stitching vias
- ▶ Sequence rails (load-switch + enable pins)
- ▶ Add UVLO/OVLO to every battery-powered design
- ▶ MLCC + bulk on both input and output
- ▶ Test with scope at SW node — look for ringing
- ▶ Measure thermal performance under worst-case load
- ▶ Document power-tree as part of schematic

Key Takeaways

Key takeaways from this lecture

LDO = quiet, SMPS = efficient

Pick by Vin/Vout ratio and noise tolerance.

Use vendor tools

WEBENCH / LTpowerCAD sizes everything for you.

Layout = success

SW loop small, ground plane continuous.

Buy the eval board

Validate ICs before designing them in.

Protect inputs

Fuse + TVS + polarity + UVLO.

Measure heat

Thermal camera + IR thermometer at full load.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ TI WEBENCH Designer • Analog Devices LTpowerCAD • MPS PowerNESS
- ▶ Robert Bosse — 'Switch-Mode Power Supplies' (textbook + lectures)
- ▶ Robert Erickson — 'Fundamentals of Power Electronics' (graduate textbook)
- ▶ Phil's Lab (YouTube) — SMPS layout & SI walkthroughs
- ▶ Robert Feranec — power-tree design videos
- ▶ Application notes from TI / Analog Devices / Monolithic Power
- ▶ Texas Instruments Power Management Lab Kit (TI-PMLK) — hands-on
- ▶ AEC-Q100/200 for automotive power IC qualification

Thank you • Build something this week.